University of Illinois **Spring 2025**

ECE 313: Problem Set 11: Problems and Solutions

Due: Monday, April 28 at 07:00 p.m.
Reading: *ECE 313 Course Notes,* Sections 4.4 - 4.8

Note on reading: For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

Note on turning in homework: Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. **Please write down your work and derivations. An answer without justification as of how it is found will not be accepted.** You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:
NAME AS IT APPEARS ON CANVAS
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SECTION
PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned page numbers.**

1. **[Sum of Continuous Variables I]**
   Suppose X and Y have the joint pdf

$$f_{X,Y}(u,v) = \begin{cases} uv & \text{if } 0 \le u \le 2,\ 0 \le v \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $S = X + Y$. Find the pdf of S, i.e., find $f_S(s)$.

Solution: In general, we have

$$f_S(s) = \int_{-\infty}^{\infty} f_{X,Y}(u, s-u)du$$

From the joint pdf, $f_{X,Y}$ is non-zero when $0 \le u \le 2$, and $0 \le s-u \le 1 \iff s-1 \le u \le s$, then our goal is to find the intersection of these two intervals for different cases of s.

- If $s < 0$, no intersection, so $f_S(s) = 0$;
- If $0 \le s < 1$, the intersection is $[0, s]$, so $f_S(s) = \int_0^s u(s-u)du = \frac{s^3}{6}$;
- If $1 \le s \le 2$, the intersection is $[s-1, s]$, so $f_S(s) = \int_{s-1}^s u(s-u)du = \frac{3s-2}{6}$;
- If $2 < s \le 3$, the intersection is $[s-1, 2]$, so $f_S(s) = \int_{s-1}^2 u(s-u)du = \frac{-s^3+15s-18}{6}$;
- If $s > 3$, no intersection, so $f_S(s) = 0$.

To sum up,

$$f_S(s) = \begin{cases} \frac{s^3}{6} & \text{if } 0 \le s < 1, \\ \frac{3s-2}{6} & \text{if } 1 \le s \le 2, \\ \frac{-s^3+15s-18}{6} & \text{if } 2 < s \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

(**Note**: Although X, Y are independent, i.e. rewriting $f_{X,Y}$ into $f_X \cdot f_Y$ is correct, it is not required since the joint pdf is directly provided.)

2. [**Sum of Continuous Variables II**]
Suppose X and Y have the joint pdf

$$f_{X,Y}(u,v) = \begin{cases} 15e^{-(2u+3v)} & \text{if } 0 \le u < v < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

Find the pdf of the sum $S = X + Y$.

Solution: Because X and Y can be any non-negative real number, we know that S must also be a non-negative real with probability 1. As a result, we can say that $f_S(s) = 0$ for $s < 0$. In general, we know that

$$f_S(s) = \int_{-\infty}^{\infty} f_{X,Y}(u, s-u)du$$

In order for $f_{X,Y} \neq 0$, it must be the case that $0 \le u < s-u < \infty$. This shows that $0 \le u < \frac{s}{2}$. Integrating along the bounds where $f_{X,Y}(u, s-u) \neq 0$ gives

$$\begin{aligned} \int_{-\infty}^{\infty} f_{X,Y}(u, s-u)du &= \int_0^{\frac{s}{2}} 15e^{-(2u+3(s-u))}du \\ &= \int_0^{\frac{s}{2}} 15e^{-(3s-u)}du \\ &= 15e^{-3s} \int_0^{\frac{s}{2}} e^u du \\ &= 15e^{-3s}(e^u)\big|_0^{\frac{s}{2}} \\ &= 15e^{-3s}(e^{\frac{s}{2}} - 1) \end{aligned}$$

Therefore

$$f_S(s) = \begin{cases} 15e^{-3s}(e^{\frac{s}{2}} - 1) & s \ge 0 \\ 0 & s < 0 \end{cases}$$

3. [**Joint PDF of Functions of Random Variables**]
Let X and Y be random variables, and define W and Z such that

$$W = \frac{1}{\sqrt{2}}(X - Y) \quad Z = \frac{1}{\sqrt{2}}(X + Y).$$

(a) If X and Y are independent $N(0,1)$ random variables, show that W and Z are also independent $N(0,1)$ random variables.

Solution: W and Z are linear mappings of X and Y, meaning that we can refer to proposition 4.7.1 in the textbook.

We know that

$$f_{W,Z}(\alpha,\beta) = \frac{1}{|\det A|} f_{X,Y}\left(A^{-1}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\right)$$

We also have the matrix equation

$$\begin{bmatrix}W\\ Z\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}X\\ Y\end{bmatrix}$$

We can find that $\det(A) = 1$ and $A^{-1} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}$ as a result. Then,

$$\begin{aligned}
f_{W,Z}(w,z) &= f_{X,Y}\left(\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}w\\ z\end{bmatrix}\right)\\
&= f_{X,Y}\left(\frac{w+z}{\sqrt{2}}, \frac{z-w}{\sqrt{2}}\right)\\
&= f_X\left(\frac{w+z}{\sqrt{2}}\right) f_Y\left(\frac{z-w}{\sqrt{2}}\right)\\
&= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(w+z)^2}{4}\right)\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(z-w)^2}{4}\right)\\
&= \frac{1}{2\pi}\exp\left(-\frac{w^2+z^2}{2}\right)\\
&= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{w^2}{2}\right)\cdot\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{z^2}{2}\right)
\end{aligned}$$

These functions are the pdfs of $N(0,1)$ random variables, meaning that W and Z are themselves $N(0,1)$ random variables.

(b) Now consider the case where X and Y are still independent, but $X \sim N(0,1)$ while $Y \sim N(0,2)$. Find the joint distribution of W and Z. Are W and Z still independent?

Solution:

$$\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(w+z)^2}{4}\right)\frac{1}{\sqrt{4\pi}}\exp\left(-\frac{(z-w)^2}{8}\right) = \frac{1}{2\pi\sqrt{2}}\exp\left(-\frac{3w^2+2wz+3z^2}{8}\right)$$

This pdf cannot be factored into functions of w and z, meaning that W and Z are not independent.

4. **[Correlation and Covariance]**

Let $E[X] = 3$, $E[Y] = 4$, $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 4$, and $\rho_{X,Y} = 0.1$.

(a) If $Z = 2(X+Y)(X-Y)$, what is $E[Z]$?

Solution:

$$\begin{aligned}E[Z] &= E[2(X+Y)(X-Y)] \\ &= 2E[X^2 - Y^2] \\ &= 2((\mathrm{Var}(X) + E[X]^2) - (\mathrm{Var}(Y) + E[Y]^2)) \\ &= 2((9+9) - (4+16)) \\ &= -4\end{aligned}$$

(b) If $T = 2X + Y$ and $U = 2X - Y$, what is $\mathrm{Cov}(T, U)$?

Solution:

$$\begin{aligned}\mathrm{Cov}(T,U) &= \mathrm{Cov}(2X+Y, 2X-Y) \\ &= \mathrm{Cov}(2X, 2X) - \mathrm{Cov}(Y, Y) \\ &= 4\,\mathrm{Var}(X) - \mathrm{Var}(Y) \\ &= 32\end{aligned}$$

(c) Find the mean and variance of $W = X + 3Y + 2$.

Solution:

$$\begin{aligned}E[W] &= E[X + 3Y + 2] \\ &= E[X] + 3E[Y] + 2 \\ &= 3 + 12 + 2 \\ &= 17\end{aligned}$$

$$\begin{aligned}\mathrm{Var}(W) &= \mathrm{Cov}(X+3Y+2, X+3Y+2) \\ &= \mathrm{Cov}(X+3Y, X+3Y) \\ &= \mathrm{Var}(X) + 9\,\mathrm{Var}(Y) + 6\,\mathrm{Cov}(X,Y) \\ &= 9 + 36 + 6(0.1\cdot 2\cdot 3) \\ &= 48.6\end{aligned}$$

5. **[Correlation and Covariance II]**
The following parts are independent.

(a) If $\mathrm{Var}(X+Y) = 36$ and $\mathrm{Var}(X-Y) = 16$, what is $\mathrm{Cov}(X,Y)$?
If you are also told that $2\,\mathrm{Var}(X) = 3\,\mathrm{Var}(Y)$, what is $\rho_{X,Y}$?

Solution:

$$\begin{aligned}&\mathrm{Var}(X+Y) = 36 \\ \implies\ &\mathrm{Cov}(X+Y, X+Y) = 36 \\ \implies\ &\mathrm{Cov}(X,X) + 2\,\mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,Y) = 36 \\ \implies\ &\mathrm{Var}(X) + 2\,\mathrm{Cov}(X,Y) + \mathrm{Var}(Y) = 36\end{aligned}$$

and

$$\begin{aligned}&\mathrm{Var}(X-Y) = 16 \\ \implies\ &\mathrm{Cov}(X-Y, X-Y) = 16 \\ \implies\ &\mathrm{Cov}(X,X) - 2\,\mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,Y) = 16 \\ \implies\ &\mathrm{Var}(X) - 2\,\mathrm{Cov}(X,Y) + \mathrm{Var}(Y) = 16\end{aligned}$$

Therefore, $\operatorname{Var}(X+Y) - \operatorname{Var}(X-Y) = 4\operatorname{Cov}(X,Y) = 20$, or

$$\operatorname{Cov}(X,Y) = 5$$

We can then also solve for $\operatorname{Var}(X)$ and $\operatorname{Var}(Y)$ and get

$$\operatorname{Var}(X) = \frac{78}{5}$$

$$\operatorname{Var}(Y) = \frac{52}{5}$$

Then,

$$\rho_{X,Y} = \frac{5}{\sqrt{\frac{78\cdot 52}{25}}} = \frac{25\sqrt{6}}{156}$$

(b) If $\operatorname{Var}(X+Y) = \operatorname{Var}(X-Y)$, are X and Y uncorrelated?
Solution: Yes, they are uncorrelated, since

$$\begin{aligned}
&\operatorname{Var}(X+Y) = \operatorname{Var}(X-Y) \\
&\implies \operatorname{Cov}(X,X) + 2\operatorname{Cov}(X,Y) + \operatorname{Cov}(Y,Y) = \operatorname{Cov}(X,X) - 2\operatorname{Cov}(X,Y) + \operatorname{Cov}(Y,Y) \\
&\implies 2\operatorname{Cov}(X,Y) = -2\operatorname{Cov}(X,Y) \\
&\implies \operatorname{Cov}(X,Y) = 0
\end{aligned}$$

(c) If $\operatorname{Var}(X) = \operatorname{Var}(Y)$, are X and Y uncorrelated?
Solution: No. A counterexample can be found by setting X to a common distribution and let $Y = X$. For example, suppose X is a Bernoulli(p) random variable where $0 < p < 1$, and let $Y = X$. Then $\operatorname{Var}(X) = \operatorname{Var}(Y)$, but $\operatorname{Cov}(X,Y) = \operatorname{Cov}(X,X) = p(1-p) \neq 0$.
You may also use other counterexamples. For example, suppose X is a Bernoulli(0.5) random variable, and let $Y = 1 - X$. Then X and Y are identically distributed, and thus $E[X] = E[Y] = 0.5$ and $\operatorname{Var}[X] = \operatorname{Var}[Y] = 0.25$. However, $E[XY] = E[X] - E[X^2] = 0$, meaning that $\operatorname{Cov}(X,Y) = E[XY] - E[X]E[Y] = -0.25 \neq 0$.